

COURSE SYLLABUS

CS 494 – Internetworking Protocol

1. GENERAL INFORMATION

Course name:	Internetworking Protocol
Course name (in Vietnamese):	Giao thức Internet
Course ID:	CS 494
Knowledge block:	
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	Introduction to Programming (C, C++, Java) or equivalent
Prior-course:	
Instructors:	Dr. Nguyen Tuan Nam

2. COURSE DESCRIPTION

The course is designed to provide students the fundamentals of computer networking, the operation of the protocols in the TCP/IP suite, Internet, LANs, packet-switching networks, network architecture. Using the Internet as a vehicle, this course introduces the underlying concepts and principles of modern computer networks, with emphasis on protocols, architectures, and implementation issues. Topics include network applications (email, ftp, http, dns, etc.), Internet in a top-down fashion: Application, Transport, Network, Data Link and Physical layers. The course will also cover some advance topics including network security and wireless networking.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
G1	Understand the basic concepts of the 5/7 Internet layers	LO1
G2	Design network protocols for network applications	LO1, LO2, LO3
G3	Develop network applications	LO1, LO2, LO3, LO4, LO6
G4	Optimize networking code for better performance	LO1, LO3, LO5
G5	Know the history of the Internet	LO1, LO11
G6	Understand the pros and cons of packet switch vs. circuit switch network	LO1, LO2

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	An ability to apply knowledge of computing and mathematics appropriate to the discipline	T
G1.2	An ability to analyze a problem, and identify and define the computing requirements appropriate to its solution	T
G1.3	An ability to design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs	T
G2.1	An ability to function effectively on teams to accomplish a common goal	I

G2.2	An understanding of professional, ethical, legal, security and social issues and responsibilities	I
G2.3	An ability to communicate effectively with a range of audiences	I
G2.4	An ability to analyze the local and global impact of computing on individuals, organizations, and society	I
G3.1	Recognition of the need for and an ability to engage in continuing professional development	T
G3.2	An ability to use current techniques, skills, and tools necessary for computing practice	T
G4.1	An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices	U
G4.2	An ability to apply design and development principles in the construction of software systems of varying complexity	U

5. TEACHING PLAN

THEORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Introduction Overview of the Internet, client/server paradigm, circuit switching, packet switching, physical media, queuing delay and packet loss	G1.1 G1.2 G1.3	Lecturing Q&A, Group discussion (Chapter 1)	QZ: Quiz MT: Midterm

2	Introduction (cont.) Overview of TCP/IP and OSI reference models, Internet Protocol Stack	G2.1 G2.2 G2.3 G2.4	Lecturing Q&A, Group discussion	QZ: Quiz MT: Midterm
3	Application Layer Service requirements, Client-server model vs. P2P model Protocols: WWW, HTTP, FTP	G3.1 G3.2	Lecturing Q&A, Group discussion	QZ: Quiz MT: Midterm
4	Application Layer (cont.) Protocols: Electronic Mail, Domain Name System, Socket programming	G4.1 G4.2	Lecturing Q&A, Group discussion	QZ: Quiz MT: Midterm
5	Transport Layer Service models, Multiplexing/Demultiplexing, Connection-less transport Principles of reliable data transfer (UDP),	G4.1 G4.2	Lecturing Q&A, Group discussion	QZ: Quiz MT: Midterm
6	Transport Layer (cont.) Connection-oriented transport (TCP), TCP congestion control, TCP Variants	G4.1 G4.2	Lecturing Q&A, Group discussion	QZ: Quiz FN: Final
7	Network Layer Two main tasks: routing and forwarding	G4.1 G4.2	Lecturing Q&A, Group discussion	QZ: Quiz FN: Final

	Forwarding in VC and Datagram networks IP protocol			
8	Network Layer (cont.) ICMP, Routing algorithms (Link State vs Dijkstra)	G4.1 G4.2	Lecturing Q&A, Group discussion	QZ: Quiz FN: Final
9	Network Layer (cont.) Routing in the Internet (AS, BGP)	G4.1 G4.2	Lecturing Q&A, Group discussion	QZ: Quiz FN: Final
10	Link Layer and Local Area Network (LAN) Link layer services, Error detection and correction, Multiple Access Protocols Link layer addressing, Ethernet, Hubs and switches, Point-to-Point Protocol	G4.1 G4.2	Lecturing Q&A, Group discussion	QZ: Quiz FN: Final
11				

** The above is a tentative schedule only. It may be updated during the course.*

6. LABORATORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Socket programming	G1.2, G1.3, G2.1, G3.2, G4.1, G4.2	Provide instructions, Demonstration, Discussion	Project 1
2	Link and Network with Software Defined Networking	G1.2, G1.3, G3.2, G4.1, G4.2	Provide instructions, Demonstration, Discussion	Project 2
3	Capture and analyze network packets	G1.2, G1.3, G3.2, G4.1, G4.2	Provide instructions, Demonstration, Discussion	Project 3

7. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Extra Credit			Up to 15%
A11	Pop Quizzes, Class Participation and/or Attendance	Small pop quizzes in class for each topic Students are randomly picked to answer question given by instructor.	G4.1 G4.2	
A2	Projects			30%
A21	Projects	Socket programming projects	G3.2	30%

A3	Exams			70%
A31	Midterm exam	Closed book exam (multiple-choice format) Describe the understanding of different topics, analyze & program to solve problems	G4.1 G4.2	20%
A33	Final exam	Closed book exam (multiple-choice format) Describe the understanding of different topics, analyze & program to solve problems	G4.1 G4.2	50%

Academic Dishonesty is not accepted in this class. You are not allowed to share your work (projects, quizzes, and exams) to others. You are not allowed to copy from your classmate's work as well. **Please read the Academic Dishonesty section for more detail of the consequences.**

The following information details the grading and examination policies. This information is subject to change at the professor's discretion:

- Pop quizzes will not be announced upfront, so make sure to attend all lectures so you won't miss any. **There will be NO make-up pop quiz** if you miss one so please make sure to attend all the lectures.
- Late Submissions/Make-up Exam Policy
 - You are not permitted to make up test or exams for this course unless documentation of an excused absence is provided. In this case, the instructor will decide if your absence is excusable to make up the exam.
 - If you miss a test or exam and have a documented excuse, you must contact the professor by email no later than the day following the missed exam or test, or you will NOT be able to take the assessment, and as a result you will receive a zero (0) for the exam.
 - Homework will be collected and checked for completion throughout the semester on due dates. Late work will not be considered and will result in a loss of points unless a documented excuse is provided.

- **NO late homework or assignments will be accepted. As a result, you will receive a zero (0) for that homework or assignment!**
- Testing Procedures:
 - You are not permitted to leave during a test or exam unless there is a documented medical situation.
 - The professor reserves the right to assign seats for tests and exams and/or move students during any classroom activity.
 - Cellular phones, personal computers, or any other wireless devices are not permitted for any test/exam. All electronic devices must be turned off and put completely away and out of sight.
 - Collaboration between students and discussion of test/exam answers are NOT permitted.
 - The use of notes, books, binder, texts, etc. is not permitted for any exam or test.
 - For **online** exams, tests, and pop quizzes:
 - **You MUST turn on your web cam during the entire time of the exam/test.** You are not allowed to turn it off until the instructor tells you to do so. **Failure to do so will result in a grade of zero (0) for the exam.**
 - The online exams will be given through FIT's Moodle. Make sure you have full access to moodle to complete the online exam.
 - Only one question will be presented at a time. You cannot go back to any previous question. **Remember to click submit before the time is over.**

8. RESOURCES

Textbooks

- Kurose and Ross. **Computer Networking: A Top-down Approach** (8th Ed. 2020 or later). Publisher: Pearson

9. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.

- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.
- You are expected to be on time for each class meeting. Attendance is taken in the first 20 minutes of each class. Any student who arrives after attendance has been taken will be marked as absence.
- For online session, if you have an emergency business and need to be away for a few minutes, **you MUST leave a note on Zoom chat box before you leave. If you leave the online lecture without sending a note on the Zoom chat box of the ongoing lecture, you will be considered as being absent without permission.**
- **Academic Honesty**
 - Your academic work must be the result of your own thought, research, or self-expression. Academic misconduct includes, but is not limited to the following: cheating, plagiarism, unacceptable collaboration, falsification of data, aiding and abetting dishonesty, unauthorized or malicious interference, hacking computer property or software, and online disturbances. Please refer to the Student Code of Conduct Policy and the Student Code of Conduct section located in the Student Handbook for more information. All required class activities are subject to submission to Turnitin.com for the detection of plagiarism. All submitted papers will be included as source documents in the Turnitin.com reference database solely for the purpose of detecting plagiarism of such papers.
- **Types of Academic Dishonesty:**
 - Cheating: Is defined as obtaining or attempting to obtain, or aiding another to obtain credit for work by dishonest or deceptive means.
 - Plagiarism: The use of words or ideas of the original creator without attribution as if they were your own. Plagiarism ranges from copying someone else's work verbatim and elaborating on or altering someone else's work.
 - Self-Plagiarism: Submitting or reusing parts of a previous paper without referencing source it was first submitted. This includes retaking a course and turning in previously submitted papers and data.

- Unacceptable Collaboration: Using answers, solutions, or ideas that are the results of collaboration without citing the fact of the collaboration is improper.
- Falsification of Data: Making up or falsifying information and data. Examples include making up or altering data for an experiment or citing reference to sources you did not actually use.
- Pay Services: Employing an assignment writing service or having another write the paper for you.
- Enabling: Aiding and abetting another student in an act of academic dishonesty. Examples include giving someone a paper to copy and allowing someone access to your account.
- Unauthorized or malicious interferences: If one person deliberately interferes with the work or activities of another person on purpose to cause the other harm or irreparable damages. Academic honesty violations are considered a breach of policy and may result in academic penalties (zero points on the assignment/test in question, and/or a failing grade for the class), disciplinary action, and/or referral to the Dean of Student Affairs.
- **Each instance of Academic Dishonesty will render a grade of 0 for the assignment/test/exam.**
- Plagiarism
 - Courses use tools such as Turnitin to deter and detect academic dishonesty. In this course, the first time you are found being academically dishonest, you will receive a zero for the assignment. If you are found academically dishonest a second time, you will receive an F (or zero) in the course. Academic dishonesty may result in further disciplinary action and/or a referral to the Dean of Student Affairs. For more information, refer to the Student Code of Conduct Policy and the Student Code of Conduct section located in the Student Handbook.